# Interpretable Machine Learning

# Feature Importance
# Permutation Feature Importance (PFI)

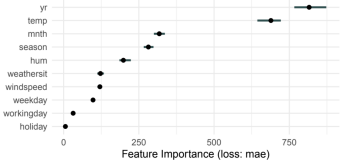
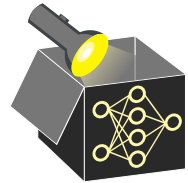



**Figure:** Bike Sharing Dataset

**Learning goals**

- Understand how PFI is computed
- Understanding strengths and weaknesses

# MOTIVATION FOR PFI



- **Goal:** Assess how important feature(s) $X_S$ are for predictive performance of a **fixed trained model** $\hat{f}$ on a given dataset $\mathcal{D}$
- **Idea:** Estimate performance change when $X_S$ is "made uninformative"

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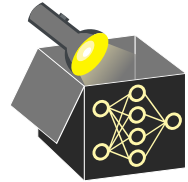

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- **Question:** Can we make $X_S$ uninformative by removing it from model?
  $\rightsquigarrow$ No, $\hat{f}$ was trained with $X_S$; retraining without $X_S$ gives a different model

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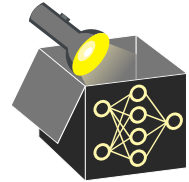

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- **Question:** Can we make $X_S$ uninformative by removing it from model?
  $\rightsquigarrow$ No, $\hat{f}$ was trained with $X_S$; retraining without $X_S$ gives a different model
- **Solution:** Simulate feature removal by replacing $X_S$ with a perturbed version $\tilde{X}_S$ that is independent of $(X_{-S}, Y)$ but preserves distrib. $\mathbb{P}(X_S)$
  $\rightsquigarrow$ Compare baseline predictions $\hat{f}(X)$ with perturbed predictions $\hat{f}(\tilde{X}_S, X_{-S})$

$$\text{PFI}_S := \underbrace{\mathbb{E}\Big[L\big(\hat{f}(\tilde{X}_S, X_{-S}), Y\big)\Big]}_{\text{risk after "destroying" } X_S} - \underbrace{\mathbb{E}\Big[L\big(\hat{f}(X), Y\big)\Big]}_{\text{baseline risk}},$$

- **How to perturb $X_S$?**
  - Add random noise: distorts $\mathbb{P}(X_S)$ (not used)
  - Permutation: preserves marginal $\mathbb{P}(X_S)$, breaks dependence with $Y$ (used)

# PERMUTATION FEATURE IMPORTANCE (PFI)

▸ "Breiman" 2001

**Sample estimator (using independent test set $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n}$)**

- Measure error **with feat. values** $x_S$ and **with permuted feat. values** $\tilde{x}_S$
- Repeat permutation (e.g., $m$ times) and average difference of both errors:

$$\widehat{PFI}_S = \frac{1}{m} \sum_{k=1}^{m} \left[ \mathcal{R}_{\text{emp}}(\hat{f}, \tilde{\mathcal{D}}_{(k)}^{S}) - \mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D}) \right]$$

  - $\mathcal{D}_S^{(k)}$: dataset with column(s) $x_S$ are **permuted** once (in repetition $k$)
  - $\mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D}) = \frac{1}{n} \sum_{(x,y) \in \mathcal{D}} L(\hat{f}(x), y)$: Measures performance of $\hat{f}$ using $\mathcal{D}$
  - Average over $m$ permutations to reduce Monte-Carlo variance

**Example** of permuting feature $x_S$ with $S = \{1\}$ and $m = 6$ permutations:

$\mathcal{D}$

| $\mathbf{x}_1$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 1 | 4 | 7 |
| 2 | 5 | 8 |
| 3 | 6 | 9 |

$\Rightarrow$

$\tilde{\mathcal{D}}_{(1)}^{S}$

| $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 1 | 4 | 7 |
| 2 | 5 | 8 |
| 3 | 6 | 9 |

$\tilde{\mathcal{D}}_{(2)}^{S}$

| $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 2 | 4 | 7 |
| 1 | 5 | 8 |
| 3 | 6 | 9 |

$\tilde{\mathcal{D}}_{(3)}^{S}$

| $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 2 | 4 | 7 |
| 3 | 5 | 8 |
| 1 | 6 | 9 |

$\tilde{\mathcal{D}}_{(4)}^{S}$

| $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 1 | 4 | 7 |
| 3 | 5 | 8 |
| 2 | 6 | 9 |

$\tilde{\mathcal{D}}_{(5)}^{S}$

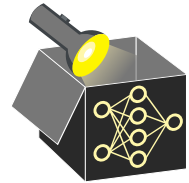
| $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 3 | 4 | 7 |
| 1 | 5 | 8 |
| 2 | 6 | 9 |

$\tilde{\mathcal{D}}_{(6)}^{S}$

| $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 3 | 4 | 7 |
| 2 | 5 | 8 |
| 1 | 6 | 9 |

Note: $S$ refers to a subset of features, here $|S| = 1$ to measure impact of permuting $x_1$ on performance

# PERMUTATION FEATURE IMPORTANCE



$\tilde{\mathcal{D}}^S_{(k)}$

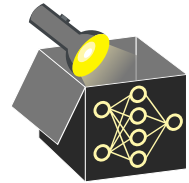
| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|---|
| 1 | 2 | 4 | 7 |
| ⋮ | 1 | 5 | 8 |
| n | 3 | 6 | 9 |

$\mathcal{D}$

| $\mathbf{x}_1$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 1 | 4 | 7 |
| 2 | 5 | 8 |
| 3 | 6 | 9 |

1. **Perturbation:** Sample feature values from the distribution of $x_S$ ($P(X_S)$).
   $\Rightarrow$ Randomly permute feature $x_S$
   $\Rightarrow$ Replace $x_S$ with permuted feat. $\tilde{x}_S$ and create data $\tilde{\mathcal{D}}^S$ containing $\tilde{x}_S$

# PERMUTATION FEATURE IMPORTANCE



$\tilde{\mathcal{D}}^S_{(k)}$

| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|---|
| 1 | 2 | 4 | 7 |
| ⋮ | 1 | 5 | 8 |
| n | 3 | 6 | 9 |

| $\hat{f}$ |
|---|
| 0.6 |
| 0.6 |
| 0.6 |

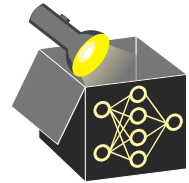
$\mathcal{D}$

| $\mathbf{x}_1$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 1 | 4 | 7 |
| 2 | 5 | 8 |
| 3 | 6 | 9 |

| $\hat{f}$ |
|---|
| 0.4 |
| 0.8 |
| 0.6 |

1. **Perturbation:** Sample feature values from the distribution of $x_S$ ($P(X_S)$).
   $\Rightarrow$ Randomly permute feature $x_S$
   $\Rightarrow$ Replace $x_S$ with permuted feat. $\tilde{x}_S$ and create data $\tilde{\mathcal{D}}^S$ containing $\tilde{x}_S$
2. **Prediction:** Make predictions for both data, i.e., $\mathcal{D}$ and $\tilde{\mathcal{D}}^S$

# PERMUTATION FEATURE IMPORTANCE



$\tilde{\mathcal{D}}^{S}_{(k)}$

| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|---|
| 1 | 2 | 4 | 7 |
| ⋮ | 1 | 5 | 8 |
| n | 3 | 6 | 9 |

| $L(\hat{f}, y)$ |
|---|
| 0.9 |
| 0.5 |
| 0.1 |

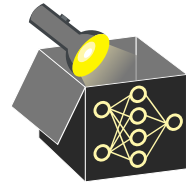
$\mathcal{D}$

| $\mathbf{x}_1$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 1 | 4 | 7 |
| 2 | 5 | 8 |
| 3 | 6 | 9 |

| $L(\hat{f}, y)$ |
|---|
| 0.25 |
| 0.35 |
| 0.1 |

3. **Aggregation:**
   - Compute the loss for each observation in both data sets

# PERMUTATION FEATURE IMPORTANCE



$\tilde{\mathcal{D}}^{S}_{(k)}$

| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|---|
| 1 | 2 | 4 | 7 |
| ⋮ | 1 | 5 | 8 |
| n | 3 | 6 | 9 |

$\mathcal{D}$

| $\mathbf{x}_1$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 1 | 4 | 7 |
| 2 | 5 | 8 |
| 3 | 6 | 9 |

| $\Delta\mathbf{L}$ |
|---|
| 0.65 |
| 0.15 |
| 0 |

| $L(\hat{f}, y)$ | | $L(\hat{f}, y)$ |
|---|---|---|
| 0.9 | | 0.25 |
| 0.5 | - | 0.35 |
| 0.1 | | 0.1 |

**3. Aggregation:**

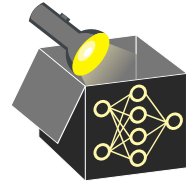
- Compute the loss for each observation in both data sets
- Take the difference of both losses $\Delta L$ for each observation

# PERMUTATION FEATURE IMPORTANCE

$$\mathcal{R}_{\text{emp}}(\hat{f}, \tilde{\mathcal{D}}^{S}_{(k)}) - \mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D})$$

| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ | $\mathbf{x}_1$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ | $\Delta\mathbf{L}$ |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 4 | 7 | 1 | 4 | 7 | 0.65 |
| ⋮ | 1 | 5 | 8 | 2 | 5 | 8 | 0.15 |
| n | 3 | 6 | 9 | 3 | 6 | 9 | 0 |

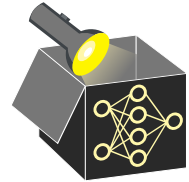
= 0.267



**3. Aggregation:**

- Compute the loss for each observation in both data sets
- Take the difference of both losses $\Delta L$ for each observation
- Average this change in loss across all observations
  Note: Same as computing $\mathcal{R}_{\text{emp}}$ on both data sets and taking difference

# PERMUTATION FEATURE IMPORTANCE



$$\mathcal{R}_{\text{emp}}(\hat{f}, \tilde{\mathcal{D}}^{S}_{(k)}) - \mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D})$$

1

| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|---|
| 1 | 2 | 4 | 7 |
| ⋮ | 1 | 5 | 8 |
| n | 3 | 6 | 9 |

| $\mathbf{x}_1$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 1 | 4 | 7 |
| 2 | 5 | 8 |
| 3 | 6 | 9 |

| $\Delta\mathbf{L}$ |
|---|
| 0.65 |
| 0.15 |
| 0 |

= 0.267

⋮

m

| $i$ | $\mathbf{x}_S$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|---|
| 1 | 3 | 4 | 7 |
| ⋮ | 2 | 5 | 8 |
| n | 1 | 6 | 9 |

| $\mathbf{x}_1$ | $\mathbf{x}_2$ | $\mathbf{x}_3$ |
|---|---|---|
| 1 | 4 | 7 |
| 2 | 5 | 8 |
| 3 | 6 | 9 |

| $\Delta\mathbf{L}$ |
|---|
| 0.85 |
| 0 |
| 0.35 |

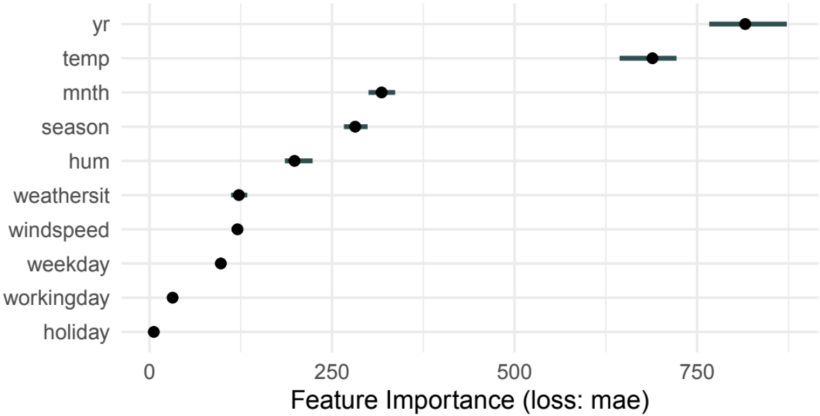
= 0.4

$\widehat{PFI}_S$ = ½ (0.267 + 0.4)

**3. Aggregation:**

- Compute the loss for each observation in both data sets
- Take the difference of both losses $\Delta L$ for each observation
- Average this change in loss across all observations
- Repeat perturbation and average over multiple repetitions

# EXAMPLE: BIKE SHARING DATASET




**Interpretation:**

- `yr` and `temp` are most important feats using mean absolute error (MAE)
- Destroying info. about `yr` by permuting it increases MAE of model by 816
- Error bars show 5% and 95% quantiles over multiple permutations

# COMMENTS ON PFI

- Interpretation: Increase in error when feature's information is destroyed

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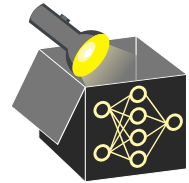

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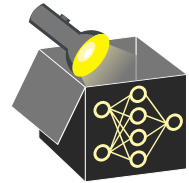

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- Permuting features despite correlation/dependence with other features can lead to unrealistic combinations of feature values
  $\rightsquigarrow$ Extrapolation issue

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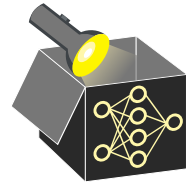

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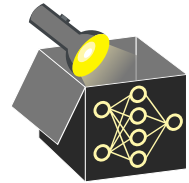
  $\rightsquigarrow$ Extrapolation issue
- PFI automatically includes importance of interaction effects with other features
  $\Rightarrow$ Permuting $x_j$ also destroys interactions with permuted feature
  $\Rightarrow$ PFI score contains importance of all interactions with permuted feature

# COMMENTS ON PFI



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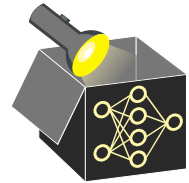
- PFI automatically includes importance of interaction effects with other features
  $\Rightarrow$ Permuting $x_j$ also destroys interactions with permuted feature
  $\Rightarrow$ PFI score contains importance of all interactions with permuted feature
- Interpretation of PFI depends on whether training or test data is used

# COMMENTS ON PFI - EXTRAPOLATION



**Example:** Let $y = x_3 + \epsilon_y$, with $\epsilon_y \sim \mathcal{N}(0, 0.1)$.

- $x_1 := \epsilon_1$, $x_2 := x_1 + \epsilon_2$; highly correlated ($\epsilon_1 \sim \mathcal{N}(0, 1)$, $\epsilon_2 \sim \mathcal{N}(0, 0.01)$)
- $x_3 := \epsilon_3$, $x_4 := \epsilon_4$, with $\epsilon_3, \epsilon_4 \sim \mathcal{N}(0, 1)$; all noise terms $\epsilon_j$ are indep.
- Fitting a linear model yields $\hat{f}(\mathbf{x}) \approx 0.3x_1 - 0.3x_2 + x_3$

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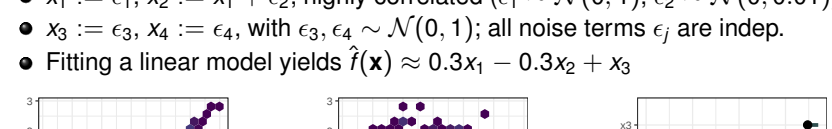


Hexbin plot of $(x_1, x_2)$ before (left) and after (center) permuting $x_1$;
PFI scores (right).

# COMMENTS ON PFI - EXTRAPOLATION

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- $x_1 := \epsilon_1$, $x_2 := x_1 + \epsilon_2$; highly correlated ($\epsilon_1 \sim \mathcal{N}(0,1)$, $\epsilon_2 \sim \mathcal{N}(0, 0.01)$)
- $x_3 := \epsilon_3$, $x_4 := \epsilon_4$, with $\epsilon_3, \epsilon_4 \sim \mathcal{N}(0,1)$; all noise terms $\epsilon_j$ are indep.
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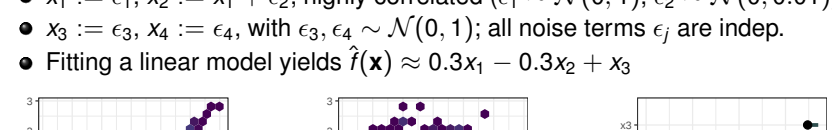


Hexbin plot of $(x_1, x_2)$ before (left) and after (center) permuting $x_1$;
PFI scores (right).

$\Rightarrow$ $x_1, x_2$ cancel in $\hat{f}$ since $x_1 \approx x_2$, hence $0.3x_1 - 0.3x_2 \approx 0$
$\rightsquigarrow$ should be irrelevant

$\Rightarrow$ Permuting $x_1$ breaks joint structure $\rightsquigarrow$ unrealistic inputs

$\Rightarrow$ $PFI > 0$ due to extrapolation (PFI evaluates model on unrealistic inputs)
$\rightsquigarrow$ $x_1, x_2$ are misleadingly considered relevant

# COMMENTS ON PFI - INTERACTIONS

**Example:** Let $x_1, \ldots, x_4$ be independently and uniformly sampled from $\{-1, 1\}$ and

$$y := x_1 x_2 + x_3 + \epsilon_Y \text{ with } \epsilon_Y \sim N(0, 1)$$

Fitting a LM yields $\hat{f}(x) \approx x_1 x_2 + x_3$.

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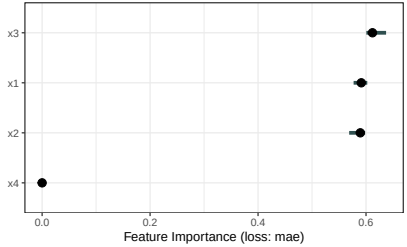
$$y := x_1 x_2 + x_3 + \epsilon_Y \text{ with } \epsilon_Y \sim N(0, 1)$$

Fitting a LM yields $\hat{f}(x) \approx x_1 x_2 + x_3$.

Although $x_3$ alone contributes as much to the prediction as $x_1$ and $x_2$ jointly, all three are considered equally relevant.

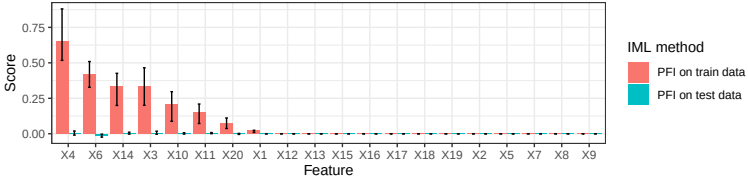
$\Rightarrow$ PFI does not fairly attribute the performance to the individual features.



# COMMENTS ON PFI - TRAIN VS. TEST DATA

**Example:**

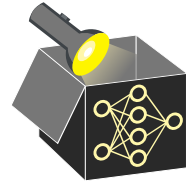
- $x_1, \ldots, x_{20}, y$ are independently sampled from $\mathcal{U}(-10, 10)$
- Train set: $n = 50$ (intentionally small) and large test set
- Model: `xgboost` with default settings (overfits strongly)




**Observation:**

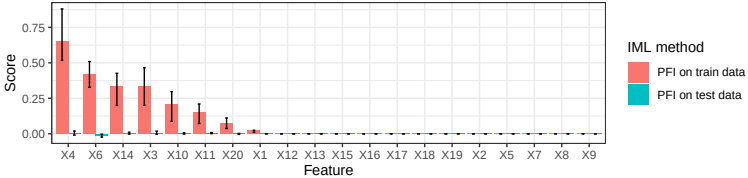
- PFI on train data highlights features that the model overfitted to.
- PFI on test data detects no relevant features.

# COMMENTS ON PFI - TRAIN VS. TEST DATA



**Example:**

- $x_1, \ldots, x_{20}, y$ are independently sampled from $\mathcal{U}(-10, 10)$
- Train set: $n = 50$ (intentionally small) and large test set
- Model: `xgboost` with default settings (overfits strongly)

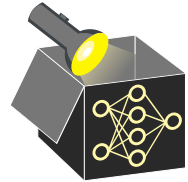



**Observation:**

- PFI on train data highlights features that the model overfitted to.
- PFI on test data detects no relevant features.

**Why?** $PFI \neq 0$ if permuting a feature breaks a dependency the model relies on. Model overfits due to spurious feature-target dependencies in train that vanish on test.
$\Rightarrow$ To find features that help the model to generalize, compute PFI on test data.

# IMPLICATIONS OF PFI



Can we get insight into whether the ...

1. feature $x_j$ is causal for the prediction?
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   - As the train vs. test data example shows, the converse does not hold

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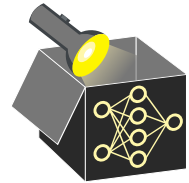

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   - $PFI_j \neq 0 \Rightarrow x_j$ is dependent on $y$, $x_{-j}$, or both (due to extrapolation)
   - $x_j$ is not exploited by model (regardless of its usefulness for $y$)
     $\Rightarrow PFI_j = 0$

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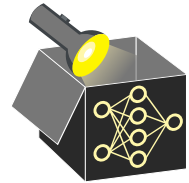

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     $\Rightarrow PFI_j = 0$
3. model requires access to $x_j$ to achieve its prediction performance?
   - As shown by the extrapolation example, such insight is not possible